Analysis of Algo.

```
i = 0          --- 1
while (i < n)  --- n+1
{
    Stmt       --- n
    i++        --- n
}
```

$T(n) = 3n + 2 \Rightarrow O(n)$

```
a = 1
while (a < b)
{
    Stmt
    a = a*2;
}
```

$O(\log_2 n) \Rightarrow (\log n)$

```
i = 1;   k = 1
while (k < n)
{
    Stmt
    k = k+i
    i++
}
```

$\Rightarrow O(\sqrt{n})$

```
while ( m != n)
{
    if (m > n){
        m = m - n }
    else
        { n = n - m }
}
```

$\Rightarrow O(m)$

m	n
16	2
14	2
12	2
10	2
8	2
6	2
4	2
2	2

$\frac{16}{2} = 8$

$\frac{m}{2} \Rightarrow m$

```
Algo Test (n){
    if (n<5)
        { stmt }          --- 1
    else {
        for ( i=0 ; i<n; i++){    --- n+1
            stmt }        --- n
    }
```

Best case $\Rightarrow O(1)$

Worst case $\Rightarrow O(n)$

$f(n) = 500$

$f(n) = 10$

$O(1) \dashrightarrow$ Constant

$f(n) = 2n + 100$

$= 500n + 10$

$O(n) \dashrightarrow$ Linear

$O(\log n) \dashrightarrow$ Logrithmic

$O(n^2) \dashrightarrow$ Quadratic

$O(n^3) \dashrightarrow$ Cubic

$O(2^n) \dashrightarrow$ Exponetial

$O(3^n)$ "

$O(4^n)$ "